

1.1.1. Calculate Momentum

07.05



Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases



calculate...

SUBMIT

```
1 m=float(input())
2 v=float(input())
3 p=m*v
4 print(F"{p:.2f}kgm/s")
```

Terminal

Test cases

< PREV RESET SUBMIT NEXT >

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1.1.2. Conditional Calculation Based on the Number of Digits06:48

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 num=int(input())
2
3 if(0<num<10):
4     print(num**2)
5
6 if(10<=num<100):
7     num1=num**0.5
8     print(f"{num1:.2f}")
9
10 if(100<=num<1000):
11     num2=num**(1/3)
12     print(f"{num2:.2f}")
13
14 if(num>=1000):
15     print("Invalid")
16
```

TerminalTest cases

< PREV

RESET

SUBMIT

NEXT >

1.1.3. Days between Two Dates

00:48



Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY – MM – DD*.
- The second line contains the second date in the format *YYYY – MM – DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases



noOfDay...

SUBMIT

```
1 from datetime import datetime
2
3 #write your code here...
4 date1 = input()
5 date2 = input()
6
7
8 d1 = datetime.strptime(date1, "%Y-%m-%d")
9 d2 = datetime.strptime(date2, "%Y-%m-%d")
10
11
12 difference = d2 - d1
13
14
15 print(difference.days)
```

Terminal

Test cases

< PREV

RESET

SUBMIT

NEXT >

1.1.4. Reverse a Number

05:04

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

+

reverseN...

SUBMIT

```
1 n=int(input())
2 num=n
3 rev=0
4 while num>0:
5
6     rev=(rev*10)+num%10
7
8     num=num//10
9 print(rev)
10
```

Terminal

Test cases

< PREVIOUS

RESET

SUBMIT

NEXT >

1.1.5. Multiplication Table

01:32



Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases



multiplica...

SUBMIT

```
1  #write your code here...
2  num = int(input())
3
4  for i in range(1,11):
5      res=i*num
6      print(f"{num} x {i} = {res}")
7
8
9
```

Terminal

Test cases

< PREV

RESET

SUBMIT

NEXT >

1.2.1. Pass or Fail

03:31

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

+

passorFa...

SUBMIT

```
1 n = int(input())
2
3 marks = list(map(int, input().split()))
4 fail = False
5 for m in marks:
6     if m < 40:
7         fail = True
8         break
9 if fail:
10    print("Fail")
11
12 else:
13    percentage = sum(marks) / n
14    print("Aggregate Percentage: {:.2f}".format(percentage))
15
16    if percentage > 75:
17        print("Grade: Distinction")
18    elif percentage >= 60:
19        print("Grade: First Division")
20
21    elif percentage >= 50:
22        print("Grade: Second Division")
23
24    elif percentage >= 40:
25        print("Grade: Third Division")
```

Terminal

Test cases

< PREV RESET SUBMIT NEXT >

1.2.2. Fibonacci Series

01:30

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

+

fibonacci....

SUBMIT

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     if n == 2:
5         return 1
6     else:
7
8         return fibonacci(n-1) + fibonacci(n-2)
9
10
11 n = int(input())
12 for i in range(1, n + 1):
13     print(fibonacci(i), end=" ")
14
```

Terminal

Test cases

< PREV

RESET

SUBMIT

NEXT >

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1.2.3. Pattern - 101:48

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n = int(input())
2
3 for i in range(1, n + 1):
4     for j in range(1, i + 1):
5         print("*", end=" ")
6     print()
```

Terminal

Test cases

< PREV

RESET

SUBMIT

NEXT >

1.2.4. Pattern - 2

01:41



Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases



numberP...

SUBMIT

```
1 # -Type- Content here...
2 n=int(input())
3 for i in range(1,n+1):
4     for j in range(1,i+1):
5         print(j,end=" ")
6     print()
7
```

Terminal

Test cases

< PREV

RESET

SUBMIT

NEXT >